

Replication and Barren Plateau Analysis of Quantum Long Short-Term Memory Networks

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1. Introduction

Quantum Machine Learning (QML) has emerged as a compelling intersection of quantum computing and classical machine learning, leveraging superposition and entanglement for pattern modelling. Among the proposed hybrid architectures, the **Quantum Long Short-Term Memory (QLSTM)** network [1] replaces parametric linear transformations inside classical LSTM cells with *Variational Quantum Circuits* (VQCs), offering theoretical expressibility advantages for sequence modelling.

However, a fundamental and underexplored challenge in all VQC-based models is the **Barren Plateau (BP) phenomenon**. The seminal survey by Larocca et al. [2] establishes that *every component* of a variational quantum algorithm—the ansatz, initial state, observable, loss function, and hardware noise—can independently cause BPs, where the optimisation landscape becomes exponentially flat and featureless as system size grows. This is not merely a theoretical concern: on a BP, an exponentially large number of measurement shots is required to identify any loss-minimising direction, making training infeasible in practice. Despite this, the trainability of QLSTM under realistic circuit configurations has not been rigorously characterised.

This project closes that gap. I perform a faithful replication of the QLSTM architecture and then conduct a systematic empirical study of BP onset within its VQC sub-circuits. Crucially, I evaluate and compare three concrete mitigation strategies, including the recent two-stage convex-initialisation framework of Boabang & Gyamerah [5], which shapes the quantum energy landscape into a smooth basin before nonconvex refinement.

The practical motivation is concrete: before investing resources in QLSTM for real-world tasks like **stock price forecasting** and **weather prediction**, a practitioner needs an evidence-based answer to—*does this model actually train, and under what circuit configurations?*

2. Project Description

The project is structured in two tightly coupled phases.

Phase 1 — Faithful Replication

I will implement the QLSTM architecture from Chen et al. [1] using PennyLane and PyTorch. Each LSTM gate’s linear transformation is replaced by a VQC sub-circuit using an angle-embedding layer followed by a strongly entangling ansatz. The primary benchmark is **stock price prediction** using daily closing prices from Yahoo Finance (e.g., S&P 500 index). Weather forecasting (Delhi or Jena Climate dataset) is included as a stretch goal if

time permits. A classical LSTM baseline with a *matched parameter count* is trained for fair comparison on the primary benchmark.

Phase 2 — Barren Plateau Analysis & Mitigation

Guided by the taxonomy in Larocca et al. [2], I identify the specific BP sources active in QLSTM (ansatz expressibility and random initialisation) and measure **gradient variance** across VQC sub-circuits as a function of:

- Circuit depth: $L \in \{1, 2, 4\}$
- Qubit count: $n \in \{2, 4, 6\}$

Gradient variance is estimated over 200 random parameter initialisations per configuration. I will then implement and compare **two** mitigation strategies:

1. **Identity-block initialisation** (Grant et al. [4]) — sub-circuits initialised so the net unitary is close to identity, keeping initial gradients non-negligible.
2. **Two-stage least squares** (Boabang & Gyamerah [5]) — a convex initialisation stage shapes the loss landscape into a smooth low-energy basin, followed by nonconvex refinement. This is the most recent (2026) mitigation framework in the literature.

AI coding agents (as permitted by the course) will be used to accelerate boilerplate circuit construction and experiment scripting.

Papers to Implement / Survey

1. Chen et al. (2020). *Quantum Long Short-Term Memory*. [arXiv:2009.01783](https://arxiv.org/abs/2009.01783) — primary architecture to replicate.
2. Larocca et al. (2024). *Barren Plateaus in Variational Quantum Computing: A Review*. [arXiv:2405.00781](https://arxiv.org/abs/2405.00781) — unified theoretical framework for BP sources and mitigations; primary reference for Phase 2 design.
3. Boabang & Gyamerah (2026). *Overcoming Barren Plateaus via a Two-Step Least Squares Approach*. [arXiv:2601.18060](https://arxiv.org/abs/2601.18060) — mitigation strategy 2; novel two-stage optimisation framework.

3. Expected Outcome

By the end of the project, I will produce:

- A working PennyLane/PyTorch QLSTM implementation reproducing the key results of Chen et al. on the stock price benchmark.
- Gradient variance plots across circuit depths $L \in \{1, 2, 4\}$ and qubit counts $n \in \{2, 4, 6\}$, characterising BP onset and cross-referenced with the taxonomy of Larocca et al. [2].
- A comparison of **two** mitigation strategies (identity-

block initialisation and two-stage least squares) with training loss curves and final MAE/RMSE.

- A quantitative QLSTM vs. classical LSTM comparison table: prediction error, parameter count, and training time.
- A clear, honest assessment of which circuit configurations make QLSTM trainable, grounded in BP theory.

4. Midterm Deliverables

- Reading notes for all three papers, with a written summary of the BP taxonomy from Larocca et al.
- Working QLSTM implementation that trains on a toy dataset (sine wave) confirms circuit wiring and gradient flow are correct.
- Classical LSTM baseline with matched parameter count on the stock price dataset.
- Preliminary gradient variance measurements at $n \in \{2, 4\}$ qubits and $L \in \{1, 2\}$ layers, confirming the experimental setup is reproducible.

5. Final Deliverables

- Full QLSTM replication on the stock price dataset with a classical LSTM comparison table (MAE, RMSE, parameter count, training time). Weather forecasting included as a stretch goal.
- Complete BP analysis: gradient variance plots across $L \in \{1, 2, 4\}$ and $n \in \{2, 4, 6\}$, annotated using BP-source taxonomy from Larocca et al.
- Comparison of both mitigation strategies (identity-block initialisation and two-stage least squares) with training loss curves and final accuracy numbers.
- Documented code repository with reproducible scripts, ready for demo during viva.
- Final report and presentation summarising all results and conclusions.

References

- [1] S. Chen, S. Yoo, and Y.-L. Fang. Quantum long short-term memory. *arXiv preprint arXiv:2009.01783*, 2020. <https://arxiv.org/abs/2009.01783>
- [2] M. Larocca, S. Thanasilp, S. Wang, K. Sharma, J. Biamonte, P. J. Coles, L. Cincio, J. R. McClean, Z. Holmes, and M. Cerezo. Barren plateaus in variational quantum computing: A review. *arXiv preprint arXiv:2405.00781*, 2024. <https://arxiv.org/abs/2405.00781>
- [3] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1):4812, 2018.
- [4] E. Grant, L. Wossnig, M. Ostaszewski, and M. Benedetti. An initialization strategy for addressing barren plateaus in parametrized quantum circuits. *Quantum*, 3:214, 2019.
- [5] F. Boabang and S. A. Gyamerah. Overcoming barren plateaus in variational quantum circuits using a two-step least squares approach. *arXiv preprint arXiv:2601.18060*, 2026. <https://arxiv.org/abs/2601.18060>